Chapter 1: The Nature of Science & Physics

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PHY201
Lecture 1

Lecture Outline

Introduction to Physics

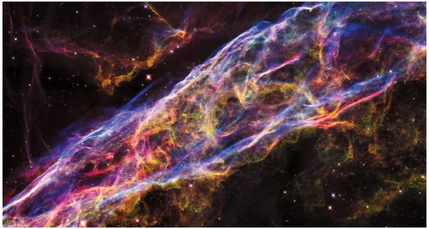
Models, Theories, and Laws

Physical Quantities & Units

Measurements

Approximation

Why Study Physics?



- It explains how everything works by connecting **space** and **time**.

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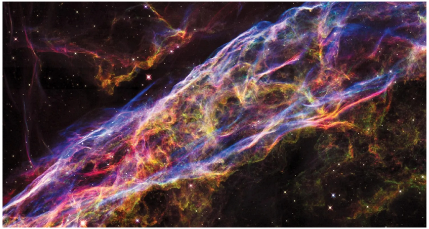

- Note: $\therefore$ means "therefore" and is used to indicate a conclusion based on the preceding statements.

- It explains how everything works by connecting **space** and **time**.
- You practice clear and logical thinking:

  Socrates is a man.

  All men are mortal.

  $\therefore$ Socrates is mortal.

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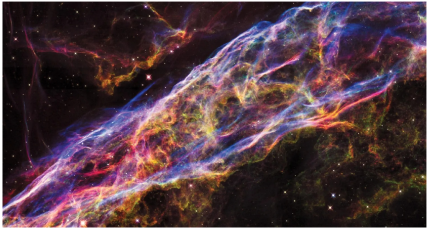

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  Socrates is a man.

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  ∴ Socrates is mortal.

- Physics concepts are used in many careers:
  - **Pilots**: how wind affects flight path
  - **Physical Therapists**: how muscles experience forces during movement

Complex World, Surprisingly Simple Laws:

The Scientific Method

The Scientific Method Uncovers the Laws of Nature

The scientific method allows us to see the underlying principles governing the behavior of the universe.

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The Scientific Method Uncovers the Laws of Nature

The scientific method allows us to see the underlying principles governing the behavior of the universe.

- Observe.
- Ask questions.
- Form hypotheses.
- Conduct experiments.
- Analyze data.
- Draw conclusions.

Physics

The Study of the Fundamental Mechanisms of the Universe

What Physics is

- Physics is the study of energy, matter, space, and time, and focuses on what fundamental mechanisms underlie every phenomenon.
- The study of physics involves breaking down complex phenomena into simpler parts.

What Physics is NOT

- Physics is not about memorizing formulas and plugging in numbers to get the "right answer"; the reasoning toward and the interpretation of this answer are crucial.

Applications of Physics

Fields that Use Physics

- **Chemistry**: interactions of atoms and molecules
- **Engineering**: applied physics
- **Architecture**: structural stability, acoustics, heating and cooling of buildings
- **Geology**: radioactive dating of rocks, earthquake analysis, and heat transfer in the Earth
- **Biophysics**: properties of cell walls and cell membranes, heat, work, and power associated with the human body, medical diagnostics, and medical therapy
- **Acoustics**: musical instruments, how sound travels through air

Lecture Outline

The Role of Experimentation: Models, Theories, and Laws

Where is Atlanta?

- Think about the city of Atlanta. How would you communicate its location to someone who has never been there?

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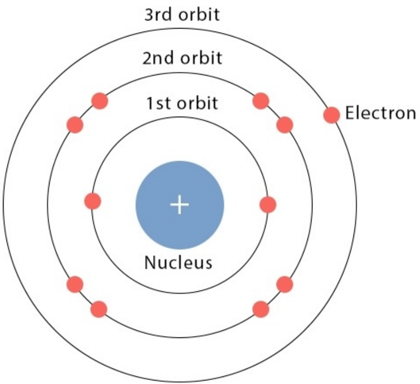
Example

- A map is a **model** of the Earth that allows us to understand and communicate the location of places.
- The map is justified with experimental proof, but it is only accurate under limited situations, such as for specific scales.

The Role of Experimentation: Models, Theories, and Laws



Definition: Model

- A **model** is a representation of something that is difficult or impossible to observe directly.

Example

- **Planetary Model of the Atom**: electrons are pictured as orbiting the nucleus similar to the way planets orbit the Sun.

The Role of Experimentation: Models, Theories, and Laws

Definition: Theory

- A **theory** is a testable explanation for patterns in nature
  - supported by scientific evidence
  - verified multiple times by various groups of researchers

Note: Theories *can* include models to help visualize phenomena.

Example

- **Newton's Theory of Gravity**: does not require a model or mental image since we can observe falling objects directly with our own senses.
- **Kinetic Theory of Gases**: requires a model in which a gas is viewed as being composed of atoms and molecules.

The Role of Experimentation: Models, Theories, and Laws

Definition: Law

- A **law** describes a generalized pattern in nature
  - supported by scientific evidence
  - verified multiple times by various groups of researchers

Difference between theory and law:
"Law" is reserved for a concise and general statement that describes phenomena in nature.
A law describes a single action, whereas a theory explains an entire group of related phenomena.

Example

$$\vec{F}_{\text{net},x} = m\vec{a}_x$$

This equation shows Isaac Newton's Second Law, which states that the net force acting on an object is proportional to the object's resulting acceleration. The larger the object's mass m is, the smaller the resulting acceleration will be for a given net force.

Check Your Understanding

Kate tells you she has just learned about a **law of nature**.

Before she describes the law to you, what can you already know about the nature of this information?

How would the information be different if Kate told you she had learned about a **scientific theory** rather than a law?

Raise your hand when you have the answer.

Lecture Outline

Physical Quantities

Definition: Physical Quantity

We define a **physical quantity** either by specifying how it is measured or by stating how it is calculated from other measurements.

- **Distance and time**: defined by specifying methods for measuring them
- **Average speed**: defined as distance traveled divided by time spent traveling

Units

Definition: Unit

A **unit** is defined as a standard in which the magnitude of a quantity is expressed.

- Length (physical quantity) can be expressed in units of inches, feet, meters, miles, etc.

Major Systems of Units

- English Units (AKA imperial): historically used in nations once ruled by the British Empire and are still widely used in the United States.
  Examples: foot, fahrenheit, pound
- Système International (SI) Units: mostly identical to the metric system.
  Examples: meter, kelvin, newton

Fundamental SI Units

Quantity	Name	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Amount of substance	mole	mol

Derived SI Units

Quantity	Name	Symbol	In Base Units	Equivalent Form
Force	newton	N	$kg \cdot m/s^2$	-
Pressure	pascal	Pa	N/m^2	J/m^3
Energy	joule	J	$N \cdot m$	$Pa \cdot m^3$
Area	square meter	m^2	m^2	-
Volume	cubic meter	m^3	m^3	-
Density	kg per cubic meter	kg/m^3	kg/m^3	-

Prefix	**Symbol**	**Value**	**Example (typical)**
tera	T	10^{12}	terawatt (TW) $\sim 10^{12}$ W — powerful laser output
giga	G	10^{9}	gigahertz (GHz) $\sim 10^{9}$ Hz — microwave frequency
mega	M	10^{6}	megacurie (MCi) $\sim 10^{6}$ Ci — high radioactivity
kilo	k	10^{3}	kilometer (km) $= 10^{3}$ m — about 0.6 mile
—	—	10^{0}	(unity)
deci	d	10^{-1}	deciliter (dL) $= 10^{-1}$ L — less than half a soda
centi	c	10^{-2}	centimeter (cm) $= 10^{-2}$ m — fingertip thickness
milli	m	10^{-3}	millimeter (mm) $= 10^{-3}$ m — flea height at shoulders
micro	µ	10^{-6}	micrometer (µm) $= 10^{-6}$ m — microscope detail
nano	n	10^{-9}	nanogram (ng) $= 10^{-9}$ g — tiny dust speck
pico	p	10^{-12}	picofarad (pF) $= 10^{-12}$ F — small radio capacitor
femto	f	10^{-15}	femtometer (fm) $= 10^{-15}$ m — proton size scale

Check Your Understanding

Humphrey the Hummingbird beats his wings 50 times per second. What is the time it takes for Humphrey to beat his wings one time?

Which unit should be used to describe the measurement?

Raise your hand when you have the answer.

Converting from One Unit to Another Unit

We can convert from unit X to unit Y by using **conversion factors**

Example

How many inches are in a mile?
Note: there are 5280 feet in a mile and 12 inches in a foot.

$$1\,\text{mi}\left(\frac{5280\,\text{ft}}{1\,\text{mi}}\right)\left(\frac{12\,\text{in}}{1\,\text{ft}}\right) = 63\,360\,\text{in}$$

$$\frac{1\cdot 5280\cdot 12}{1\cdot 1}\left(\frac{\text{mi}\cdot\text{ft}\cdot\text{in}}{\text{mi}\cdot\text{ft}}\right) = 63\,360\,\text{in}$$

See that $\left(\frac{5280\,\text{ft}}{1\,\text{mi}}\right)$ and $\left(\frac{12\,\text{in}}{1\,\text{ft}}\right)$ are conversion factors.

$\because$ 5280 feet and 1 mile represent the same distance, $\left(\frac{5280\,\text{ft}}{1\,\text{mi}}\right) = 1$.

Practice

Unit Conversions (cp2e_ch1_ex1)

Suppose that you drive the 10.0 km from your school to home in 20.0 min. Calculate your average speed (a) in kilometers per hour (km/h) and (b) in meters per second (m/s).

(Solution is shown in lecture)

Practice

Unit Conversions (cp2e_ch1_p4)

American football is played on a 100-yd-long field, excluding the end zones. How long is the field in meters? (Assume that 1 meter equals 3.28 feet.)

(Solution is shown in lecture)

Practice

Unit Conversions

A dish contains 200 mL of water. What is this volume in cubic centimers?

Note: $1000\,\mathrm{L} = 1\,\mathrm{m}^3$

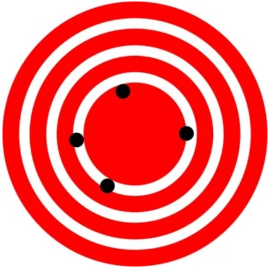
(Solution is shown in lecture)

Lecture Outline

Importance of Measurements

Science is based on observation and experiment - that is, on **measurements**.

Accuracy of a Measurement



Accuracy is how close a measurement is to the correct value for that measurement.

Precision of a Measurement

The **precision** of a measurement system refers to how close the agreement is between repeated measurements (which are repeated under the same conditions).

Certainty in Measurements

Some measurements convey more confidence in their accuracy than others.

Which do you expect to be closer to 5 kilograms?:

- 5-kg Bag of Potatoes
- 5-kg Weight Plate used in an Olympic Weightlifting Competition

Significant Figures

We communicate this certainty by using significant figures.

- Mass of Potato Bag: 5 kg
- Mass of Weight Plate: 5.00 kg $\longleftarrow$ these extra zeroes are *significant*

Significant figures communicate how much certainty there is in measured values.

Significant Figures

1. **Rules for Counting Significant Figures** ***(sf or sig. figs)***
   - All non-zero digits are significant.
     (e.g. 123 has 3 sf)
   - Zeros between non-zero digits are significant.
     (e.g. 12304 has 5 sf)
   - Leading zeros are not significant.
     (e.g. 0.0012 has 2 sf)
   - Trailing zeros are significant only if a decimal point is present.
     (e.g. 120 has 2 sf; 120. has 3 sf; 120.30 has 5 sf)

Check Your Understanding

Determine the number of significant figures in the following measurements:

a. 0.0009
b. $15,450.0$
c. 6×10^3
d. 87.990
e. 30.42

Raise your hand when you have the answer.

Significant Figures

2. Addition & Subtraction

- Round the result to the same decimal place as the **least precise** term.

  Keep the minimum number of digits after the decimal place in your answer.

$$12.11 + 0.3 + 1.025 = 13.435 \xrightarrow{\text{round to tenths}} 13.4$$

Check Your Understanding

Anna has two bags of oranges and one bag of apples.

Each bag of oranges weighs 13.52 lb, while the bag of apples weighs 10.2 lb.

What is the total weight of the bags?

Raise your hand when you have the answer.

Significant Figures

3. Multiplication & Division

- Round the result to the same *total significant figures* as the factor with the fewest.

$$4.56 \times 1.4 = 6.384 \xrightarrow{\text{round to 2 sf}} 6.4$$

Check Your Understanding

The force F on an object is equal to its mass m multiplied by its acceleration a.

$$F = ma$$

If a wagon with mass 55 kg accelerates at a rate of $0.0255\,\frac{\mathrm{m}}{\mathrm{s}^2}$, what is the force on the wagon?

(The unit of force is called the newton (symbol: N))

Raise your hand when you have the answer.

Example: Significant Figures

Considering Significant Figures in Calculations (cp2e_ch1_p22)

What is the area of a circle 3.102 cm in diameter?

$$A_\circ = \pi r^2$$

$$A_\circ = \pi \left(\frac{d}{2}\right)^2$$

$$A_\circ = \pi \left(\frac{3.102\,\text{cm}}{2}\right)^2$$

$$A_\circ = 7.557\,418\,429\,1\,\text{cm}^2$$

$$A_\circ = 7.557\,\text{cm}^2$$

1. Count the significant figures (sf).
   there are 4 sf in 3.102 cm
2. Apply sf rules for multiplication.
   $\therefore$ answer must have 4 sf.
3. Calculate answer to many digits.
4. Round to correct number of sf.

Percent Uncertainty of a Measurement

One method of expressing uncertainty is as a **percent of the measured value.**

If a measurement A is expressed with uncertainty δA, the percent uncertainty %unc is defined to be

$$\%\text{unc} = \frac{\delta A}{A} \times 100\%.$$

Example: Percent Uncertainty

Calculating Percent Uncertainty (cp2e_ch1_ex2)

A grocery store sells 5-lb bags of apples.
You purchase four bags over the course of a month and weigh the apples each time.
You obtain the following weight measurements per week: 4.8 lb, 5.3 lb, 4.9 lb, 5.4 lb.

You have previously determined, using additional measurements, that the uncertainty in the average weight of the 5-lb bag is ± 0.4 lb.

What is the **average weight** and **percent uncertainty** of the bag's weight, given these 4 measurements?

Average weight:

$$\frac{1}{4}(4.8 + 5.3 + 4.9 + 5.4)\,\text{lb} = \boxed{5.1\,\text{lb}}$$

$$\%\text{unc} = \frac{\delta A}{A} \times 100\%$$

$$\%\text{unc} = \frac{0.4\,\text{lb}}{5.1\,\text{lb}} \times 100\% = \boxed{8\%}$$

Example: Percent Uncertainty

$\therefore$ we can express the weight of a 5-lb bag of apples in terms of:

absolute certainty: $5.1\,\text{lb} \pm 0.4\,\text{lb}$

OR

percent uncertainty: $5.1\,\text{lb} \pm 8\%$

Check Your Understanding

A high-school track team's top sprinter clocked a 100-meter sprint at $12.04\,\mathrm{s}$ last week and at $11.96\,\mathrm{s}$ this week.

The manual of the stopwatch used states that the stopwatch has an uncertainty of $\pm 0.05\,\mathrm{s}$.

Can we conclude that the sprinter improved their time?

Raise your hand when you have the answer.

Lecture Outline

Use What You Know

Scientists and engineers are often expected to solve problems with limited information.

"I don't know" is not a satisfactory answer.

"I don't know, but I can estimate it" is better.

Prove Your Skills in Problem-Solving and Communication

Many employers use approximation problems in interviews to assess how candidates think under uncertainty rather than what they know.

These questions require breaking a complex problem into smaller parts, identifying key factors, and making clear, defensible assumptions.

The goal is to combine these pieces into a logical estimate. The answer matters less than how the candidate structures, reasons, and clearly explains the solution.

Fields that Gauge Approximation Ability

- Consulting (McKinsey, Deloitte, Accenture)
- Software Engineering (Google, Amazon, Meta, Microsoft)
- Investment Banking & Equity Research (Goldman Sachs, JPMorgan Chase)

Approximation is a Skill

As you develop your problem-solving skills, you will also develop skills at approximating.

Your approximating ability will improve with **practice** and as you become more **familiar with units**.

When approximating, there are many valid ways to arrive at an answer. As long as your way is logical, you are approximating successfully.

Example: Approximation

How many individual pieces of cereal does all of America eat in a week?

Questions to ask and assumptions to make:

- How many pieces of cereal are in a bag of cereal?

$$\approx 3000 \text{ pieces}$$

- How many bowls of cereal does the average person in America eat per week?

$$\approx 3.0 \text{ bowls}$$

- How many bowls of cereal are in the average bag of cereal?

$$\approx 4 \text{ bowls}$$

- How many people live in America?

$$\approx 350 \times 10^6 \text{ persons}$$

Example: Approximation

How many individual pieces of cereal does all of America eat in a week?

Synthesize your assumptions and solve.

$$350 \times 10^6 \text{ persons} \left(\frac{\left(\frac{3.0 \text{ bowls}}{1 \text{ person}}\right)}{\text{week}} \right) \left(\frac{1 \text{ bag}}{4 \text{ bowls}} \right) \left(\frac{3000 \text{ pieces}}{1 \text{ bag}} \right) = 7.875 \times 10^{11} \text{ pieces/week}$$

$$\approx \boxed{800 \text{ billion pieces/week}}$$

Notice: since there is only 1 sf in “4 bowls of cereal per bag of cereal”, and sf rules when multiplying/dividing state that we keep the minimum number of sf, our final answer must have 1 sf.

Example: Approximation

How many individual pieces of cereal does all of America eat in a week?

Questions to ask and assumptions to make:

- How much cereal does America eat per year? (searched)

$$\approx 6.4 \times 10^9 \text{ lb/year}$$

- What is the mass of the average piece of cereal?

$$\approx 0.07 \text{ g/piece}$$

- How many grams are in a pound?

$$453.6 \text{ g/lb}$$

- How many weeks are in a year?

$$52 \text{ weeks/year}$$

Example: Approximation

How many individual pieces of cereal does all of America eat in a week?

Synthesize your assumptions and solve.

$$6.4 \times 10^9 \ \frac{\text{lb}}{\text{year}} \left(\frac{453.6 \text{ g}}{1 \text{ lb}}\right)\left(\frac{1 \text{ piece}}{0.07 \text{ g}}\right)\left(\frac{1 \text{ year}}{52 \text{ weeks}}\right) = 7.975 \times 10^{11} \text{ pieces/week}$$

$$\approx \boxed{800 \text{ billion pieces/week}}$$

Since this alternative and independent method arrives at the same estimate as our original method, we are highly confident that our estimate is reasonable.

Additional Problems for Consideration

- How many gas stations are there in the world?
- How many steps have you taken in your lifetime?
- How many crosswords have been solved in history?
- How many times has the average person read, heard, or said the word “the”?
- How much total time have humans talked on their phones?
- How many plastic flamingos are there in the US?
- How many hours of your life have you lost to doomscrolling?
- How many computers are in the state?